Robotics for Juniors

Engineering with constraints: sensors, competition rules and a team that ships.

Robotique Juniors

NEXT COHORT	SCHEDULE	DURATION	SEATS LEFT
18 April 2026	Saturday 13:00–16:00	12 weeks	4

WHAT YOU WILL BUILD

- Design a robot around a defined challenge and rule set.
- Integrate multiple sensors and explain the signal flow.
- Work in a team with clear build, test and review roles.
- Document and pitch an engineering decision.

SYLLABUS OUTLINE

01 Challenge brief
Constraints, scoring and strategy: what winning actually requires.

02 Mechanical design
Chassis, gearing and torque decisions made against the rule set.

03 Sensor integration
Combine distance, line and touch sensors and trace the signal flow.

04 Control logic
State machines and calibration for repeatable autonomous runs.

05 Competition build
Iterate the build under time pressure, with test data driving changes.

06 Final run
Run the challenge, present the engineering log and review the result.

CAPSTONE & STUDENT OUTPUT

A competition-ready robot

A rule-compliant robot, an engineering log and a short design defence presented to mentors and families.

STUDENT OUTPUT
An autonomous sensor-driven robot, documented code and a team demo.